(6) $\lim_{x\to 2}\dfrac{\frac{1}{x}-\frac{1}{2}}{x-2}$ (7) $\lim_{x\to 1}\dfrac{\sqrt{x}-x^2}{1-\sqrt{x}}$ (8) $\lim_{x\to 0}\dfrac{\sqrt{x^2+1}-1}{\sqrt{x^2+16}-4}$

(9) $\lim_{x\to 0}\dfrac{\sqrt{1+x}-1}{x}$ (10) $\lim_{x\to 1}\dfrac{\sqrt[3]{7+x^3}-\sqrt{3+x^2}}{x-1}$ (11) $\lim_{x\to 2}\dfrac{2-\sqrt{x+2}}{\sqrt[3]{2}-\sqrt[3]{4-x}}$

(12) $\lim_{x\to 0}\dfrac{\sqrt{1+x^2}-1}{x}$ (13) $\lim_{x\to 0}\dfrac{\sqrt{1-x}-1}{x^2}$ (14) $\lim_{x\to 5}\dfrac{\sqrt{x-1}-2}{x-5}$

(15) $\lim_{x\to a}\dfrac{\sqrt{x-b}-\sqrt{a-b}}{x^2-a^2}(a>b)$

9.2.4 Infinite limits and limits at infinity

Infinite Limits



Let $f:\mathbb{R}\setminus\{0\}\to\mathbb{R}$ be defined by $f(x)=\dfrac{1}{x^2}$.

Let us consider the problem of calculating $\lim_{x\to 0}\dfrac{1}{x^2}$.

The following table gives the values of $\dfrac{1}{x^2}$ near 0.

$x\to 0^+$			$x\to 0^-$		
x	x^2	$\frac{1}{x^2}$	x	x^2	$\frac{1}{x^2}$
1	1	1	−1	1	1
0.5	0.25	4	− 0.5	0.25	4
0.1	0.01	100	− 0.1	0.01	100
0.01	0.0001	10,000	− .01	0.0001	10,000
0.001	0.000001	10,00,000	− 0.001	0.000001	10,00,000
0.0001	0.00000001	10,00,00,000	− 0.0001	0.00000001	10,00,00,000

The table values tell us that as x gets closer and closer to 0, $f(x)=\dfrac{1}{x^2}$ gets larger and larger. In fact, $\dfrac{1}{x^2}$ grows without bound as x approaches 0 from either side. In this situation we say that $f(x)$ **tends to infinity as x approaches zero** and write $\dfrac{1}{x^2}\to\infty$ as $x\to 0^-$ and $\dfrac{1}{x^2}\to\infty$ as $x\to 0^+$

and hence $\dfrac{1}{x^2}\to\infty$ as $x\to 0$.

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